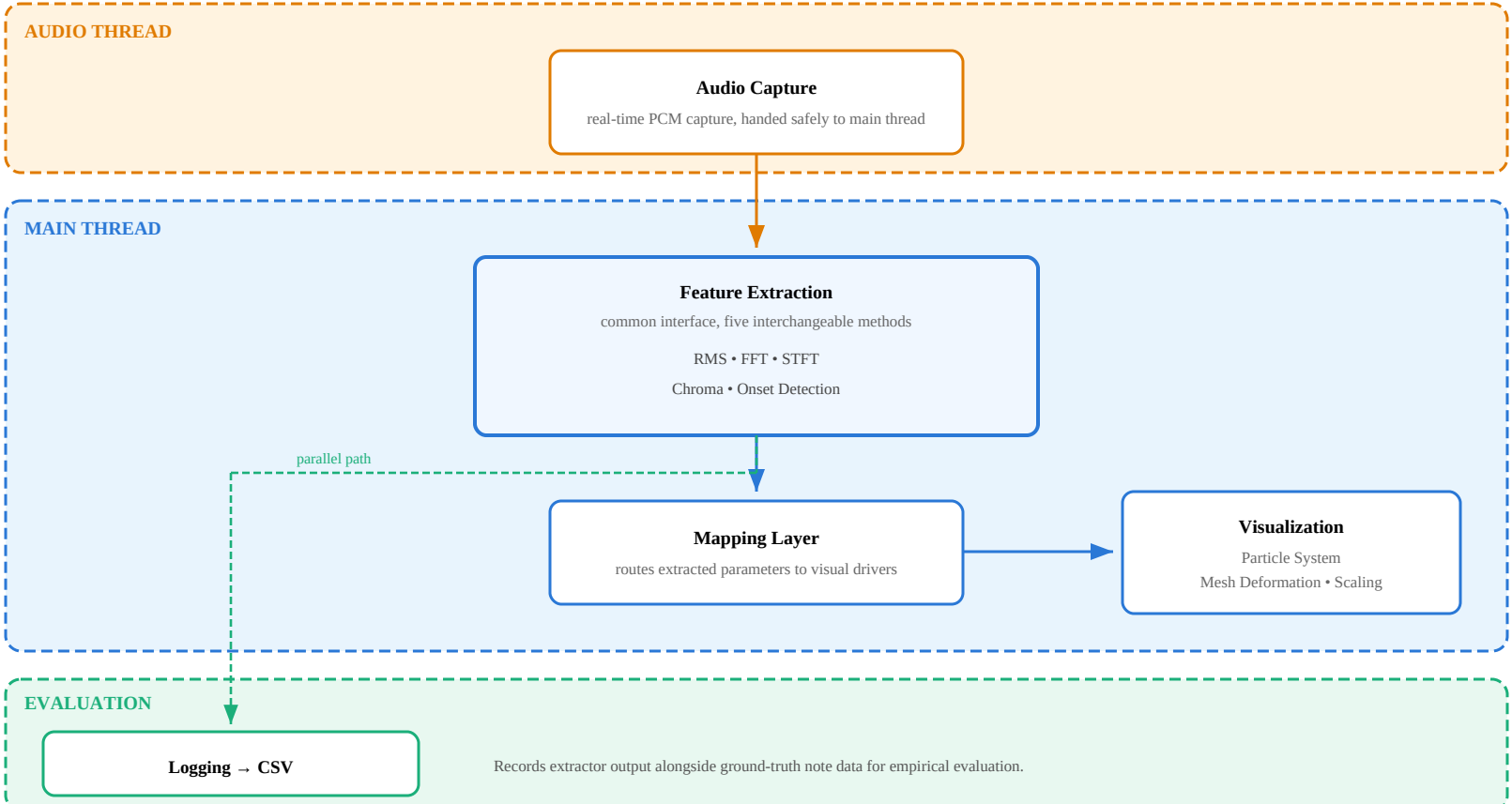
AUDIO THREAD
Audio Capture
real-time PCM capture, handed safely to main thread
MAIN THREAD
Feature Extraction
common interface, five interchangeable methods
RMS • FFT • STFT
Chroma • Onset Detection
parallel path
Mapping Layer
routes extracted parameters to visual drivers
Visualization
Particle System
Mesh Deformation • Scaling
EVALUATION
Logging → CSV
Records extractor output alongside ground-truth note data for empirical evaluation.